MH1301 Discrete Mathematics

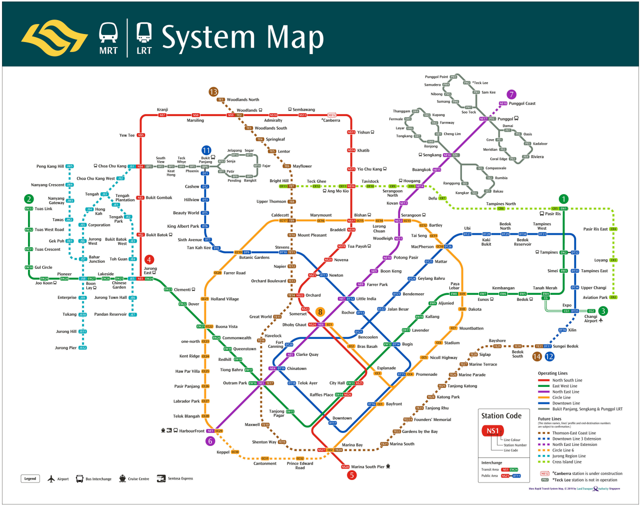
Handout 7: Graph Theory (1)

Synopsis

We begin our discussion on graph theory, which is part 3 of this course. In this handout, we are mainly introducing terminologies and definitions. We also cover a basic result which is sometimes called the Handshaking Lemma.

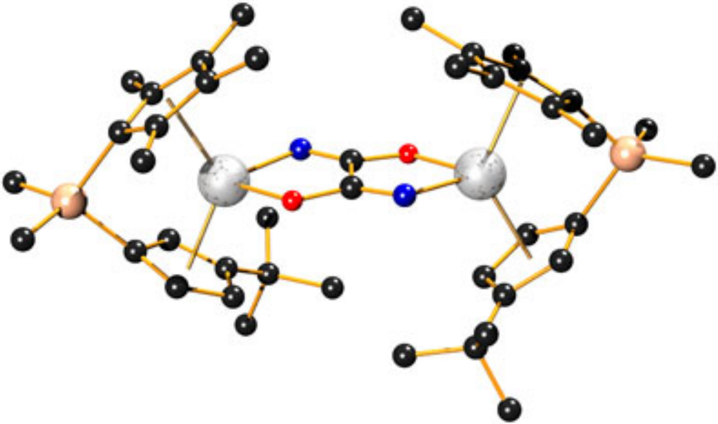
- Graph and graph models
- Graph terminology and special types of graphs
- Graph isomorphism

MRT

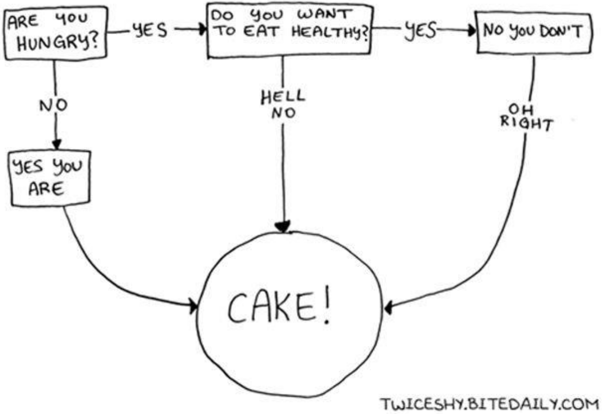

MRT
LRT
System Map
Station Code
NS1
Line Colour
Station Number
Line Code
Interchange
Transit Area
Public Area
Operating Lines
North South Line
East West Line
North East Line
Circle Line
Downtown Line
Bukit Panjang, Sengkang & Punggol LRT
Future Lines
Thomson-East Coast Line
Downtown Line 3 Extension
North East Line Extension
Circle Line 6
Jurong Region Line
Cross Island Line
*Canberra station is under construction
*Teck Lee station is not in operation
Legend
Airport
Bus Interchange
Cruise Centre
Sentosa Express

Facebook

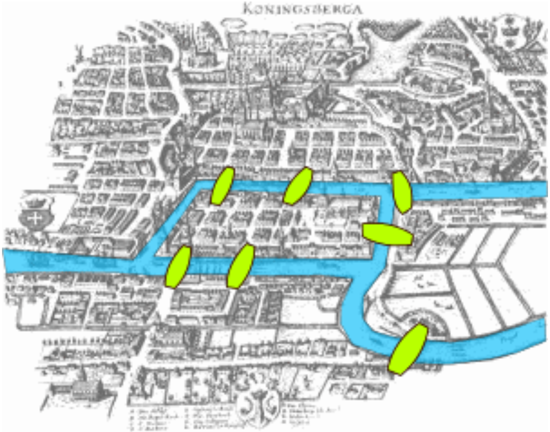
Chemical Bonds



Flowchart


ARE YOU HUNGRY?
YES
DO YOU WANT TO EAT HEALTHY?
YES
NO YOU DON'T
NO
HELL NO
OH RIGHT
YES YOU ARE
CAKE!
TWICESHY.BITEDAILY.COM

The Bridges of Königsberg



In 1735, Euler wondered if he can have a walk that traverses each of the 7 bridges exactly once

What's in Common

- Each of these structures consists of
  - a collection of objects and
  - links between those objects.

- **Goal:** find a general framework for describing these objects and their properties.

- A graph is a mathematical structure for representing relationships.

Graphs

- Graphs consist of vertices and edges .
  – Sometimes also called nodes and arcs. → not used in this course
  nodes ↓ vertices, arcs ↓ edges
- An edge connects two vertices.
- Edges can be directed or undirected ,
  - i.e., point from A to B or just connect A and B.
- **Graph Theory** studies graphs and their properties.

Formalizing Graphs

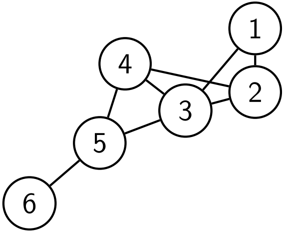
Definition 1 (Section 10.1)

vertex singular

A graph $G = (V, E)$ consists of a nonempty set V of vertices (or nodes) and a set E of edges.

plural

Each edge has either one or two vertices associated with it called its endpoints. An edge is said to connect its endpoints.



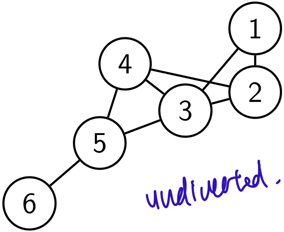
if edge only have one associated vertex, it connects to itself

Formalizing Graphs

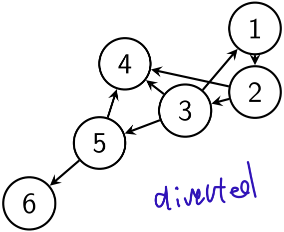
basically the lines.

Definition 2 (Section 10.1)

- A directed graph (or digraph) (V, E) consists of a nonemphty set of vertices V and a set of directed edges E. Each directed edge, $e \in E$, is associated with an **ordered** pair (u, v) for some $u, v \in V$ and is said to start at u and end at v.
- Compare with an undirected graph: Each edge $e \in E$ is an **unordered** pair $\{u, v\}$ for some $u \neq v \in V$. → no starting or ending point



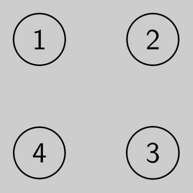
undirected.



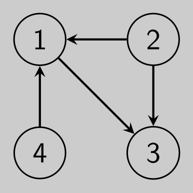
directed

Example

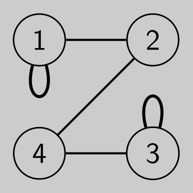
Which of the following drawings are valid **undirected** graphs?



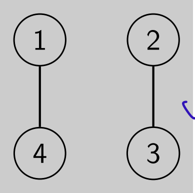
✓ vacuously true



directed graph



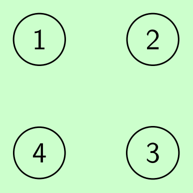
look@ previous slide - $u \neq v$ for undirected graph

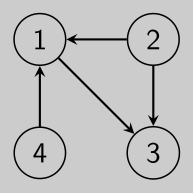


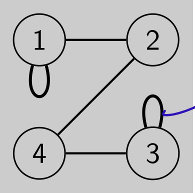
✓

Example

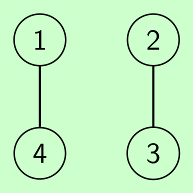
Which of the following drawings are of valid **undirected** graphs?



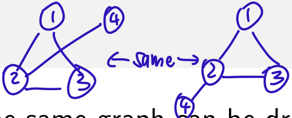




self-loops not allowed!
↓
for undirected graphs



Notes



- The same graph can be drawn in different ways, i.e., the exact location of vertices in the drawing etc., is **not** part of the graph.

- We usually denote $n = |V|$ as the number of vertices and $m = |E|$ as the number of edges.

- An undirected graph is also called a simple graph if
  - edges of the form $\{v_1, v_1\}$ are not allowed (no self-loops),
  - there can be at most one edge between any two vertices, and
  - $0 \leq m \leq C(n, 2)$.

- In a multigraph, self-loops and multiple edges between same pair of vertices are allowed.

Unless specified otherwise, all graphs in the lectures, assignments, and exams are simple graphs.

Neighbors and Incidence

Definition 1 (Section 10.2)

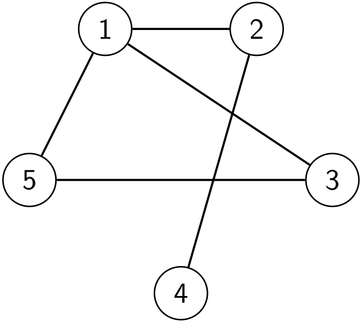
Two vertices $u, v \in V$ of a graph $G = (V, E)$ are adjacent (or neighbors) if there is an edge $\{u, v\} \in E$, i.e. if there is an edge connecting u and v.

directly connects that two points

If $e = \{u, v\} \in E$, then e is incident with u and v, and both u and v are endpoints of e.

Example

Which pairs of vertices are adjacent?

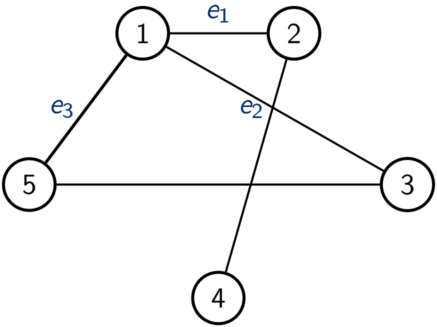


Solution:

- 1 and 5
- 1 and 3
- 1 and 2
- 2 and 4
- 3 and 5

Example

- Which vertices are edge e_1 incident with?
- Which vertices are edge e_2 incident with?
- Which edges are Vertex 1 endpoint of?



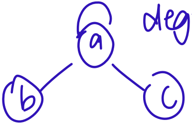
Solution:

- e_1 is incident with 1 and 2.
- e_2 is incident with 1 and 3.
- Vertex 1 is endpoint of e_1, e_2 and e_3.

Degree

Definition 3 (Section 10.2)

- The degree of a vertex $v \in V$ is the number of edges incident to v. The degree is denoted by $deg(v)$.
- The degree of a graph $G = (V, E)$, denoted by $deg(G)$ is the **maximum** degree of any vertex $v \in V$.

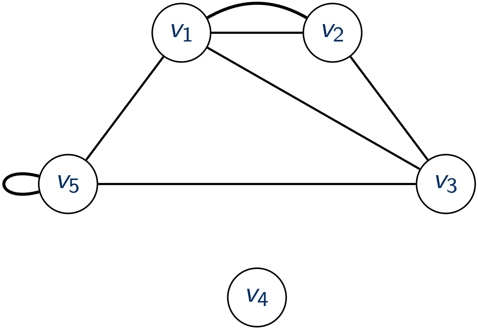


degree of a is 4

- A vertex of degree 0 is isolated.
- If (for a multigraph) a vertex v has a self-loop, then this edge contributes 2 to the degree of v.
- If all vertices have the same degree k, then the graph is called k-regular.

Example

- What are the degrees of the vertices?
- Which vertices are isolated?



Solution:

- $deg(v_1) = 4$
- $deg(v_2) = 3$
- $deg(v_3) = 3$
- $deg(v_4) = 0$
- $deg(v_5) = 4$
- v_4 is isolated

Handshaking Lemma

Theorem 1 (Section 10.2)

Let $G = (V, E)$ be a graph with m edges. Then

$$2m = \sum_{v \in V} deg(v).$$

Proof: Each edge contributes two to the degree count of all vertices, since each edge is incident to two vertices.

Hence, the sum of degrees of all vertices equals twice the number of edges. $\square$

Example

Example 3 (Section 10.2)

How many edges are there in a 6-regular graph with 10 vertices?

all vertices have same degree of 6.

Solution:

- Let m be the number of edges in the graph.
- By the Handshaking Lemma, we know $2m = 10 \cdot 6$.
- Hence $m = 30$.

Handshaking Lemma, II

Theorem 2 (Section 10.2)

In every graph G, there is an even number of vertices of odd degree.

Proof: Let V_0 be the set of vertices with even degree and V_1 the set of vertices of odd degree. Then

$$\underbrace{2m}_{\text{even}} = \sum_{v \in V} deg(v) = \underbrace{\sum_{v \in V_0} deg(v)}_{\text{even}} + \underbrace{\sum_{v \in V_1} deg(v)}_{\text{also even!}}$$

Hence $\sum_{v \in V_1} deg(v)$ is even.

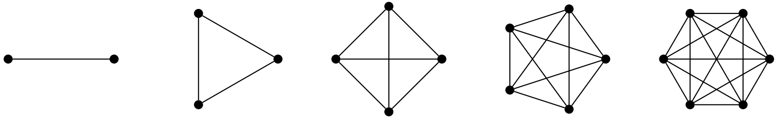
This is only possible if $|V_1|$ is even. $\square$

Special Graphs

Complete Graph (Example 5 Section 10.2)

A complete graph on n vertices, denoted by K_n, is the (unique) graph that contains all possible edges, i.e., every pair of vertices is connected by an edge.

- The complete graph has degree $n-1$.
- Here are the first few complete graph: (K_1 is just a single vertex)

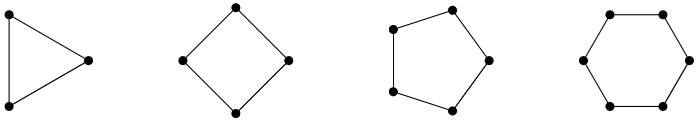


Special Graphs

Cycle (Example 6 Section 10.2)

A cycle graph, denoted by C_n on $n \geq 3$ vertices $v_1, \ldots, v_n$, has edge set $\{\{v_1, v_2\}, \{v_2, v_3\}, \{v_3, v_4\}, \ldots, \{v_{n-1}, v_n\}, \{v_n, v_1\}\}$.

- All cycles have degree 2.
- Here are the first few cycles C_3 to C_6:

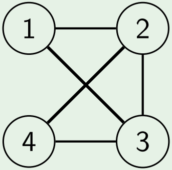


Subgraphs

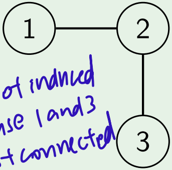
Definitions 7 and 8 (Section 10.2)

- A subgraph of a graph $G = (V, E)$ is a graph $H = (W, F)$ such that $W \subseteq V$ and $F \subseteq E$.
- $H = (W, F)$ is an induced subgraph of $G = (V, E)$, if F contains **all** edges from E that connect vertices in W.

Example

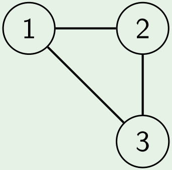


Graph G



not induced
cause 1 and 3
not connected

A subgraph of G



An induced sugraph of G

Graph Isomorphism

Definition 1 (Section 10.3)

Two (undirected) graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic, if there is a bijection function $f : V_1 \to V_2$, such that for all $u, v \in V_1$:

$$\{u, v\} \in E_1 \Longleftrightarrow \{f(u), f(v)\} \in E_2.$$

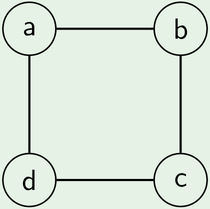
- Intuitively, isomorphic graphs are "THE SAME", except for "re-named" vertices.
- Two graphs that are not isomorphic are called non-isomorphic.

Example of isomorphic graphs

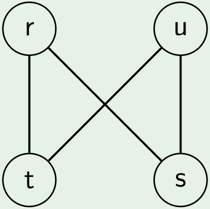
$f: V \to W$

Example 8 (Section 10.3)

Graph $G = (V, E)$ and $H = (W, F)$ are isomorphic.



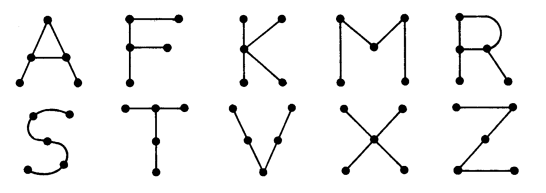
Graph G



Graph H

The function f with $f(a) = r$, $f(b) = s$, $f(c) = u$ and $f(d) = t$ is a one-to-one correspondence between V and W.

Example – Letters



Which of the above "letters" are isomorphic? (From the book S.Lipschutz and M.Lipson. Schaum's Outline of Discrete Mathematics, McGraw Hill 2007, Revised 3rd edition.)

Solution:

- A - R
- F - T
- K - X
- M - Z, S, V

Isomorphism

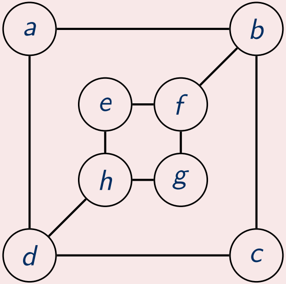
- It is difficult to determine whether two graphs are isomorphic by bruce force: there are $n!$ bijections between vertices of two n-vertex graphs.

- Till today, we still don't know whether there is an efficient algorithm/procedure for this task.

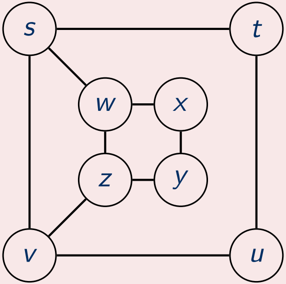
- Often, one can show two graphs are **not** isomorphic by finding a property that only one of the two graphs has. Such a property is called graph invariant, e.g.
  - number of vertices of given degree
  - the degree sequence
  - ...

Examples

Are these graphs isomorphic?



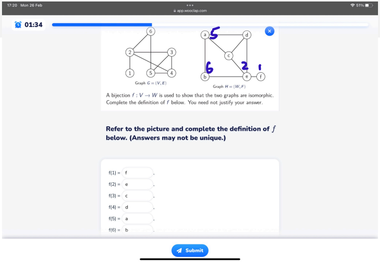
Graph G



Graph H

Solution: No! Since $deg(a) = 2$ in G, a must correspond to t, u, x or y, since these are the vertices of degree 2 in H. But a is adjacent to 2 other vertices of degree 3, which is not true of any of those vertices in H.

Reading Assignment



- Rosen (7th edition):
  - Chapter 10.1. Graphs and graph models
  - Chapter 10.2. Graph terminology and special types of graphs
  - Chapter 10.3. Graph isomorphism